

CSS

Complete Reference Guide

[Links](#) · [Lists](#) · [Tables](#) · [Display](#) · [Position](#) · [Flexbox](#) · [Grid](#) · [Forms & More](#)

CSS Links	CSS Lists	CSS Tables	CSS Display
CSS Max-width	CSS Position	CSS Overflow	CSS Float
CSS Inline-block	CSS Align	CSS Combinators	CSS Pseudo-class
CSS Pseudo-element	CSS Opacity	CSS Navigation Bar	CSS Dropdowns
CSS Image Gallery	CSS Image Sprites	CSS Attr Selectors	CSS Forms
CSS Counters	CSS Backgrounds	CSS Borders	CSS Margins
CSS Padding	CSS Height/Width	CSS Box Model	CSS Outline
CSS Text	CSS Fonts	CSS Icons	CSS Colors

Table of Contents

1. Backgrounds
2. Borders & Outline
3. Box Model (Margins · Padding · Height/Width)
4. Text & Fonts
5. Colors & Opacity
6. Display & Visibility
7. Position & Overflow
8. Float & Inline-block
9. Flexbox & Alignment
10. Max-width
11. Links
12. Lists
13. Tables
14. Forms
15. Navigation Bar & Dropdowns
16. Combinators & Selectors
17. Pseudo-classes & Pseudo-elements
18. Attribute Selectors
19. Counters
20. Image Gallery & Sprites
21. Icons
22. CSS Variables (Custom Properties)
23. Quick-Reference Cheat Sheet

1. CSS Backgrounds

CSS background properties control the background of HTML elements — color, image, position, size, and repetition.

Key Properties

Property	Description	Example Value
background-color	Solid fill color	##ff0000, rgba(0,0,0,0.5)
background-image	Image or gradient	url('img.png'), linear-gradient(...)
background-repeat	Tile behavior	no-repeat repeat-x repeat-y
background-position	Image starting point	center top 50% 50%
background-size	Scale image	cover contain 100px 200px
background-attachment	Scroll behavior	fixed scroll local
background-clip	Painting area	border-box padding-box content-box
background-origin	Origin of position	border-box padding-box content-box

```
div { background-color: #1a1a2e; background-image: url('pattern.png');  
background-repeat: no-repeat; background-position: center center; background-size:  
cover; /* Shorthand */ background: #1a1a2e url('hero.jpg') no-repeat center/cover; }
```

■ *background-size: cover scales the image to fill the element; contain scales to fit without cropping.*

2. CSS Borders & Outline

Border Properties

```
/* Individual sides */ div { border-top: 2px solid #4a9eff; border-right: 1px dashed  
red; border-bottom: 3px dotted green; border-left: 0; /* Shorthand: width style color */  
border: 2px solid #ccc; } /* Rounded corners */ button { border-radius: 8px; } .circle {  
border-radius: 50%; } /* Shorthand border-radius: top-left top-right bottom-right  
bottom-left */ .pill { border-radius: 100px 100px 100px 100px; }
```

Outline vs Border

Outline does NOT affect layout (no space taken). Border does.

```
input:focus { outline: 2px solid #4a9eff; outline-offset: 3px; /* gap between element  
edge and outline */ } /* Remove default outline (always replace with a visible focus  
style) */ button { outline: none; }
```

■ *Border styles: solid | dashed | dotted | double | groove | ridge | inset | outset | none*

■ 3. CSS Box Model — Margins · Padding · Height/Width

Every element is a rectangular box. From inside out: content → padding → border → margin.

```
/* Box sizing */ * { box-sizing: border-box; } /* width includes padding + border – recommended */ div { width: 300px; height: 150px; padding: 20px; /* all sides */ padding: 10px 20px; /* top/bottom left/right */ padding: 5px 10px 15px 20px; /* top right bottom left */ margin: 0 auto; /* center block element horizontally */ border: 1px solid #ccc; } /* Margin collapse: vertical margins of adjacent blocks merge */ p { margin-bottom: 20px; } h2 { margin-top: 30px; } /* Gap between p and h2 = 30px (not 50px) */
```

Max & Min Dimensions

```
img { max-width: 100%; /* never overflow container */ height: auto; /* maintain aspect ratio */ } .container { max-width: 1200px; min-height: 100vh; width: 90%; margin: 0 auto; }
```

■ Use *max-width* instead of *width* for responsive containers — elements shrink on small screens.

■ 4. CSS Text & Fonts

Text Formatting

```
p { color: #cdd6f4; font-size: 16px; /* or rem, em, vw */ font-weight: bold; /* 100–900 or named */ font-style: italic; text-align: center; /* left | right | center | justify */ text-decoration: underline; /* none | underline | line-through | overline */ text-transform: uppercase; /* lowercase | capitalize */ letter-spacing: 0.05em; line-height: 1.6; /* unitless recommended */ text-indent: 2em; word-spacing: 4px; white-space: nowrap; /* pre | pre-wrap | normal */ text-shadow: 2px 2px 4px rgba(0,0,0,0.5); text-overflow: ellipsis; /* requires overflow:hidden + white-space:nowrap */ overflow: hidden; }
```

Font Family & @font-face

```
/* Google Fonts import */ @import url('https://fonts.googleapis.com/css2?family=Inter:wght@400;700&display=swap'); body { font-family: 'Inter', Arial, sans-serif; } /* Custom font */ @font-face { font-family: 'MyFont'; src: url('myfont.woff2') format('woff2'); font-weight: normal; font-style: normal; } /* Generic families: serif, sans-serif, monospace, cursive, fantasy */
```

■ Always include web-safe fallback fonts and a generic family as the last value.

■ 5. CSS Colors & Opacity

```
/* Color formats */ .el { color: red; /* named color */ color: #ff6b6b; /* hex */ color: #f6b; /* shorthand hex */ color: rgb(255, 107, 107); /* RGB */ color: rgba(255, 107, 107, 0.8); /* RGBA with alpha */ color: hsl(0, 100%, 71%); /* HSL */ color: hsla(0, 100%, 71%, 0.5); /* HSLA */ /* Opacity: 0 = invisible, 1 = fully visible */ opacity: 0.7; /* affects element AND children */ /* Better alternative for just background transparency */ background-color: rgba(0, 0, 0, 0.5); } /* CSS Custom Properties (Variables) */ :root { --primary: #4a9eff; --text: #cdd6f4; --bg: #1e1e2e; } h1 { color: var(--primary); }
```

■ *opacity affects the whole element including children. Use rgba/hsla for transparent backgrounds only.*

6. CSS Display & Visibility

```
/* display values */ .block { display: block; } /* full width, new line */ .inline { display: inline; } /* flows with text, no width/height */ .inline-block { display: inline-block; } /* inline but respects width/height */ .flex { display: flex; } /* flexbox container */ .grid { display: grid; } /* grid container */ .none { display: none; } /* removed from layout */ /* Visibility: hidden hides but keeps space */ .hidden { visibility: hidden; } /* Inline-block use case */ nav a { display: inline-block; padding: 10px 16px; /* padding works! unlike pure inline */ width: 120px; /* width works! */ } /* Key differences */ /* display:none – element gone, no space */ /* visibility:hidden – invisible, space preserved */ /* opacity:0 – invisible, space preserved, still interactive */
```

7. CSS Position & Overflow

Position Values

```
/* static (default) – normal flow, top/left/etc ignored */ .static { position: static; } /* relative – offset from normal position, children anchor to this */ .relative { position: relative; top: 10px; left: 20px; } /* absolute – removed from flow, positioned relative to nearest non-static ancestor */ .tooltip { position: absolute; top: 100%; left: 0; } /* fixed – relative to viewport, stays on scroll */ .navbar { position: fixed; top: 0; width: 100%; z-index: 1000; } /* sticky – scrolls normally then sticks at threshold */ .sticky-header { position: sticky; top: 0; background: #1e1e2e; }
```

Overflow

```
div { overflow: visible; /* default – content spills out */ overflow: hidden; /* clips content */ overflow: scroll; /* always shows scrollbars */ overflow: auto; /* scrollbars only when needed */ /* Axis-specific */ overflow-x: hidden; overflow-y: auto; }
```

■ *z-index only works on positioned elements (not static). Higher value = on top.*

8. CSS Float & Inline-block

```
/* Float – element pulled left or right, text wraps around it */ img { float: left; margin-right: 16px; } /* Clearfix – parent collapse fix */ .container::after { content:
```

```

'; display: table; clear: both; } /* Or modern: */ .container { overflow: hidden; } /*
Clear – prevent wrapping on that element */ .footer { clear: both; } /* Inline-block
layout (pre-flexbox columns) */ .col { display: inline-block; width: 30%;
vertical-align: top; /* Note: whitespace between tags creates small gaps */ }

```

■ *Prefer Flexbox or Grid over floats for layout. Use floats for text wrapping around images.*

■ 9. CSS Flexbox & Alignment

```

/* Container */ .flex-container { display: flex; flex-direction: row; /* row |
row-reverse | column | column-reverse */ flex-wrap: wrap; /* nowrap | wrap |
wrap-reverse */ justify-content: space-between; /* flex-start | flex-end | center |
space-between | space-around | space-evenly */ align-items: center; /* stretch |
flex-start | flex-end | center | baseline */ align-content: flex-start; /* (multi-line)
same values as justify-content */ gap: 16px; /* gap: row-gap column-gap */ } /* Items */
.flex-item { flex: 1; /* shorthand: flex-grow flex-shrink flex-basis */ flex-grow: 2; /*
take 2x extra space */ flex-shrink: 0; /* don't shrink */ flex-basis: 200px; /* initial
size */ align-self: flex-end; /* override align-items for this item */ order: -1; /*
visual order (default 0) */ } /* Center anything */ .parent { display: flex;
justify-content: center; align-items: center; }

```

■ 10. CSS Links

Links have five states, each targetable with pseudo-classes. Order matters: LoVe HAtE (Link, Visited, Hover, Active).

```

a:link { color: #4a9eff; text-decoration: none; } a:visited { color: #cba6f7; } a:hover
{ color: #89b4fa; text-decoration: underline; } a:focus { outline: 2px solid #4a9eff;
outline-offset: 2px; } a:active { color: #f38ba8; } /* Button-style link */ a.btn {
display: inline-block; padding: 10px 24px; background: #4a9eff; color: white;
border-radius: 6px; transition: background 0.2s ease; } a.btn:hover { background:
#3d8ce8; }

```

■ 11. CSS Lists

```

ul { list-style-type: disc; /* disc | circle | square | none */ list-style-image:
url('bullet.png'); list-style-position: inside; /* inside | outside */ /* Shorthand */
list-style: square inside; padding-left: 0; margin: 0; } ol { list-style-type: decimal;
} /* decimal | lower-alpha | upper-roman etc. */ /* Reset list style */ ul.clean {
list-style: none; padding: 0; margin: 0; } /* Custom bullet with ::before */ ul.custom
li::before { content: '•'; color: #4a9eff; font-size: 0.7em; margin-right: 8px; }

```

■ 12. CSS Tables

```
table { width: 100%; border-collapse: collapse; /* merge borders */ border-spacing: 0;
/* space between cells (if separate) */ table-layout: fixed; /* equal column widths */ }
th, td { padding: 12px 16px; text-align: left; border-bottom: 1px solid #45475a; } th {
background: #313244; color: #89b4fa; font-weight: bold; } /* Zebra striping */
tr:nth-child(even) { background: #1e1e2e; } tr:nth-child(odd) { background: #181825; }
/* Hover highlight */ tbody tr:hover { background: #313244; } /* Responsive table */
.table-wrap { overflow-x: auto; }
```

■ 13. CSS Forms

```
/* Input styling */ input[type="text"], input[type="email"], textarea, select { width:
100%; padding: 10px 14px; border: 1px solid #45475a; border-radius: 6px; background:
#181825; color: #cdd6f4; font-size: 14px; outline: none; transition: border-color 0.2s;
} input:focus, textarea:focus, select:focus { border-color: #4a9eff; box-shadow: 0 0 0
3px rgba(74,158,255,0.2); } input::placeholder { color: #585b70; } /* Checkbox / Radio
*/ input[type="checkbox"], input[type="radio"] { accent-color: #4a9eff; /* modern
browser tinting */ width: 16px; height: 16px; } /* Button */ button[type="submit"] {
background: #4a9eff; color: white; border: none; padding: 10px 24px; border-radius:
6px; cursor: pointer; font-size: 14px; } button:disabled { opacity: 0.5; cursor:
not-allowed; }
```

■ 14. CSS Navigation Bar & Dropdowns

```
/* Horizontal Navbar */ nav ul { display: flex; list-style: none; margin: 0; padding: 0;
background: #181825; } nav ul li a { display: block; padding: 14px 20px; color: #cdd6f4;
text-decoration: none; transition: background 0.2s; } nav ul li a:hover { background:
#313244; color: #4a9eff; } /* Dropdown */ .dropdown { position: relative; }
.dropdown-menu { display: none; position: absolute; top: 100%; left: 0; background:
#1e1e2e; min-width: 160px; box-shadow: 0 4px 16px rgba(0,0,0,0.4); border-radius: 6px;
z-index: 100; } .dropdown:hover .dropdown-menu { display: block; } .dropdown-menu a {
display: block; padding: 10px 16px; color: #cdd6f4; } .dropdown-menu a:hover {
background: #313244; }
```

■ 15. CSS Combinators & Selectors

```
/* Descendant – any level deep */ div p { color: red; } /* Child – direct children only
*/ ul > li { font-weight: bold; } /* Adjacent sibling – immediately after */ h2 + p {
font-size: 1.1em; } /* General sibling – all siblings after */ h2 ~ p { color: gray; }
/* Attribute selectors */ a[href] { color: blue; } /* has attribute */
input[type="email"] { border: red; } /* exact match */ a[href^="https"] { color: green;
} /* starts with */ a[href$=".pdf"] { color: red; } /* ends with */ a[href*="example"] {
color: orange; } /* contains */ [class~="highlight"] { background: yellow; }/*
space-separated word */ [lang|=“en”] { font: serif; } /* exact or followed by - */
```

■ 16. Pseudo-classes & Pseudo-elements

Pseudo-classes (: single colon)

```
a:hover { } /* mouse over */ input:focus { } /* keyboard/click focus */ input:disabled { } /* disabled form element */ input:checked { } /* checked checkbox/radio */ input:required { } /* required field */ p:first-child { } /* first sibling */ p:last-child { } /* last sibling */ li:nth-child(2) { } /* 2nd child */ li:nth-child(odd){ } /* odd items */ li:nth-child(3n) { } /* every 3rd item */ p:not(.special) { } /* negation */ :root { } /* document root (html) */
```

Pseudo-elements (:: double colon)

```
p::first-line { } /* first line of text */ p::first-letter { font-size: 2em; float: left; } .item::before { content: '★ '; color: gold; } .item::after { content: ' →'; } ::selection { background: #4a9eff; color: white; } input::placeholder { color: #888; }
```

■ *::before and ::after require the content property — even empty string content."*

■ 17. CSS Counters

```
/* CSS counters — auto-numbering without JavaScript */ ol.custom { counter-reset: my-counter; /* create/reset counter */ list-style: none; } ol.custom li { counter-increment: my-counter; /* increment each item */ } ol.custom li::before { content: counter(my-counter) ' '; font-weight: bold; color: #4a9eff; } /* Nested counters */ ol { counter-reset: section; } li { counter-increment: section; } li::before { content: counters(section, '.') ' '; /* e.g. 1.2.3 */ }
```

■ 18. CSS Image Gallery & Sprites

Responsive Image Gallery

```
.gallery { display: grid; grid-template-columns: repeat(auto-fill, minmax(200px, 1fr)); gap: 12px; } .gallery img { width: 100%; height: 200px; object-fit: cover; /* cover | contain | fill | none | scale-down */ border-radius: 8px; transition: transform 0.3s; } .gallery img:hover { transform: scale(1.04); }
```

CSS Sprites

```
/* Combine multiple icons in one image to reduce HTTP requests */ .icon { background-image: url('sprites.png'); display: inline-block; } .icon-home { background-position: 0 0; width: 24px; height: 24px; } .icon-user { background-position: -24px 0; width: 24px; height: 24px; } .icon-gear { background-position: -48px 0; width: 24px; height: 24px; }
```

■ 19. CSS Icons

Use icon libraries (Font Awesome, Bootstrap Icons, Material Icons) or SVG icons.

```
/* Font Awesome 6 (CDN link in HTML head) */ /* <link rel='stylesheet'
href='https://cdnjs.cloudflare.com/...'/> */ <i class="fa-solid
fa-house"></i> /* Sizing and coloring icons */ .fa-house { font-size: 24px;
color: #4a9eff; } /* Inline SVG icon */ <svg width="24" height="24" viewBox="0 0 24
24" fill="none"> <path d="M3 12l9-9 9 9" stroke="#4a9eff" stroke-width="2">
</svg> /* Styling SVG icons via CSS */ svg { width: 24px; height: 24px; fill:
currentColor; }
```

■ 20. CSS Variables (Custom Properties)

```
:root { /* Define variables */ --color-primary: #4a9eff; --color-bg: #1e1e2e;
--color-text: #cdd6f4; --spacing-sm: 8px; --spacing-md: 16px; --spacing-lg: 32px;
--radius: 8px; --transition: 0.2s ease; } /* Use variables */ .card { background:
var(--color-bg); color: var(--color-text); padding: var(--spacing-md); border-radius:
var(--radius); } /* Fallback value */ .btn { color: var(--color-primary, #007bff); } /*
Override in component scope */ .dark-card { --color-bg: #181825; } /* Override with JS
*/ /* document.documentElement.style.setProperty('--color-primary', '#f00'); */
```

■ CSS Quick-Reference Cheat Sheet

Category	Property / Selector	Common Values
Box Model	box-sizing	border-box content-box
	margin / padding	0 auto 10px 10px 20px
	width / height	100% auto 300px 50vw
	display	block inline inline-block flex grid none
	visibility	visible hidden
	justify-content	flex-start center space-between space-evenly
	align-items	stretch center flex-start flex-end
	flex	1 0 0 auto 1 1 200px
	grid-template-columns	repeat(3, 1fr) auto-fill minmax(200px, 1fr)
	gap	16px 10px 20px
	position	static relative absolute fixed sticky
	top/right/bottom/left	0 50% auto
	z-index	0 10 999 -1
Text	font-size	16px 1rem 1.2em clamp(14px, 2vw, 18px)
	font-weight	400 700 bold lighter
	text-align	left center right justify
	line-height	1.5 1.6 24px
Selectors	Descendant	div p
	Child	ul > li
	Adjacent	h2 + p
Pseudo	Attribute	input[type='text'] a[href^='https']
	:hover :focus :nth-child(n)	State-based styling
	::before ::after	content: '' required
Transitions	transition	all 0.3s ease color 0.2s
	transform	scale(1.1) translateX(20px) rotate(45deg)
Responsive	@media	(max-width: 768px) { ... }
	min/max-width	max-width: 1200px min-width: 320px

Variables

--name: value

var(--name, fallback)

◆ CSS Complete Reference Guide — All major topics covered